

Plus, cross, even and odd

Polarization of gravitational waves reflected by a thin shell and by a black hole,
and the relative importance of the two parities

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Abstract

We study how the two polarizations h_+ and $h_\times$ of a gravitational wave are affected by a spherical compact object: a thin shell with flat interior and Schwarzschild exterior, compared with a Schwarzschild black hole. The two gauge-invariant master functions, even (Zerilli–Moncrief) and odd (Cunningham–Price–Moncrief), are not the two polarizations. They are E - and B -type patterns that both contribute to h_+ and $h_\times$, related by a 45° rotation of the polarization, and they carry energy with equal weight. Spherical symmetry and parity forbid any mixing between them. A pure- $h_\times$ axisymmetric wave therefore always comes back as pure $h_\times$. A general wave does change polarization, because the object reflects the two parities with different complex amplitudes: it behaves as a birefringent mirror. For a black hole the Chandrasekhar transformation gives this “retardance” in closed form, $\Phi = 2 \arctan[12M\omega/(\ell-1)\ell(\ell+1)(\ell+2)]$. It implies a universal long-wavelength helicity-flip cross section of exactly $4\pi M^2/3$, which we recover from a partial-wave sum. A shell reflects all the energy in both parities but with different phases. Near its cavity, trapped and matter resonances the polarization conversion becomes complete. In the time domain, the same incident pulse returns with visibly different waveforms in the two parities: only the even one rings at the shell’s matter frequency and triggers the instability of the static shell. We end with a practical answer to the question of which parity matters.

Short answers

1. **Even/odd vs $+/ \times$.** Each parity produces both polarizations. The odd field is the even field rotated by 45° in polarization. Only for axisymmetric ($m = 0$) modes is “even = h_+ , odd = $h_\times$ ” literally true.
2. **Can $h_\times$ become h_+ on reflection?** Not for a wave of pure parity: parity is exactly conserved by any spherically symmetric object, so an axisymmetric $h_\times$ wave returns as pure $h_\times$. A wave with both parities does change polarization, because $S_{\text{even}} \neq S_{\text{odd}}$. The object is a birefringent mirror, even a black hole.
3. **Which parity matters more?** In energy neither: equal weights, equal content in a plane wave, and equal reflected energy for a non-absorbing object. In physics the even sector is the richer one: it alone couples to the shell’s matter (resonances, instability). For slow sources the even sector dominates the emission, and the odd one is suppressed by $\sim v^2/36$ in power.

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1 Setting and plan

The object is a static thin shell of radius $R > 2M$ with flat interior and Schwarzschild exterior. It is supported by a surface pressure $p = \sigma(1 - \sqrt{f_R})/4\sqrt{f_R}$ with $f_R = 1 - 2M/R$, the only static configuration (Ipser and Sikivie [5]; Poisson [6], Ch. 3). Its perturbations are described outside by the Regge–Wheeler [1] and Zerilli [2] equations for the master functions of Martel and Poisson [4], and inside by the flat wave equation at the blueshifted frequency $\nu = \omega/\sqrt{f_R}$. The shell enters through the linearized Israel junction conditions [7]:

- odd parity: Ψ continuous and $[\partial_x \Psi] = \Delta \Psi$, a δ -barrier of strength $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$ in the tortoise coordinate x ;
- even parity: a frequency-dependent 2×2 transfer matrix, which includes the radial motion of the shell and the compression of its fluid (adiabatic index $\Gamma = 2$).

The even sector contains “matter modes”, one of which is unstable [8]. A Schwarzschild black hole [3, 11] is used as a reference throughout.

The plan follows the questions in the box above: Sec. 2 relates $(\Psi_{\text{even}}, \Psi_{\text{odd}})$ to $(h_+, h_\times)$; Sec. 3 derives the selection rule and the “birefringent mirror” description; Sec. 4 solves the black-hole case exactly; Sec. 5 computes the retardance and the polarization conversion of shells; Sec. 6 computes the helicity-flip cross section of a plane wave; Sec. 7 shows reflected waveforms of the same incident h_+ pulse in the two parities; Sec. 8 answers which parity matters. Units $G = c = 1$; time dependence $e^{-i\omega t}$; frequencies in units of $1/M$.

2 From the master functions to h_+ and $h_\times$

Question

The perturbation theory is written for two master functions, Ψ_{even} and Ψ_{odd} . A detector measures two polarizations, h_+ and $h_\times$. How are the two pairs related? Is “even” the same as “plus”?

2.1 The radiative field

Far from the object, in an outgoing radiation gauge, the transverse–traceless field is [4] (Eqs. 6.13–6.15)

$$h_+ \equiv \frac{p_{\theta\theta}}{r^2} = \frac{1}{r} \sum_{\ell m} \left\{ \Psi_{\text{even}}^{\ell m} W^{\ell m} - \Psi_{\text{odd}}^{\ell m} X^{\ell m} \right\}, \quad (2.1)$$

$$h_\times \equiv \frac{p_{\theta\phi}}{r^2 \sin \theta} = \frac{1}{r} \sum_{\ell m} \left\{ \Psi_{\text{even}}^{\ell m} X^{\ell m} + \Psi_{\text{odd}}^{\ell m} W^{\ell m} \right\}, \quad (2.2)$$

with the two angular functions

$$W^{\ell m} = \left[\partial_\theta^2 + \frac{1}{2}\ell(\ell+1) \right] Y^{\ell m}, \quad X^{\ell m} = \frac{im}{\sin \theta} \left[\partial_\theta - \cot \theta \right] Y^{\ell m}. \quad (2.3)$$

Here Ψ_{even} is the Zerilli–Moncrief function and Ψ_{odd} the Cunningham–Price–Moncrief function, both evaluated at $r \rightarrow \infty$ as functions of retarded time $u = t - r_*$. Combining the two equations,

$$h_+ - i h_\times = \frac{1}{2r} \sum_{\ell m} \sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \left(\Psi_{\text{even}}^{\ell m} - i \Psi_{\text{odd}}^{\ell m} \right) {}_{-2}Y^{\ell m}(\theta, \phi) \quad (2.4)$$

where ${}_{-2}Y^{\ell m}$ are the spin-weighted harmonics of weight -2 [12]. We checked the identity $W^{\ell m} - iX^{\ell m} = \frac{1}{2}\sqrt{(\ell+2)!/(\ell-2)!} {}_{-2}Y^{\ell m}$ symbolically for several (ℓ, m) . The energy and angular-

momentum fluxes are [4] (Eq. 6.16)

$$\left\langle \frac{dE}{du} \right\rangle = \frac{1}{64\pi} \sum_{\ell m} \frac{(\ell+2)!}{(\ell-2)!} \left\langle |\dot{\Psi}_{\text{even}}^{\ell m}|^2 + |\dot{\Psi}_{\text{odd}}^{\ell m}|^2 \right\rangle. \quad (2.5)$$

2.2 Four consequences

1. **Parity is not polarization.** Every (ℓ, m) mode of either parity produces, in general, *both* h_+ and $h_\times$.
2. **Duality: odd is even rotated by 45° .** Equations (2.1)–(2.2) show that an odd mode with the same Ψ as an even mode gives $(h_+, h_\times)^{\text{odd}} = (-h_\times, h_+)^{\text{even}}$. In Eq. (2.4) this is the factor $-i$, i.e. a rotation of the polarization frame by 45° . In CMB language the even field is an E -mode (gradient pattern) and the odd field a B -mode (curl pattern).
3. **Axisymmetric modes ($m = 0$) have $X^{\ell 0} = 0$:** the even mode is *pure* h_+ and the odd mode *pure* $h_\times$. Only in this case, and on the equator for $m = \pm\ell$, does “even = plus, odd = cross” hold literally.
4. **Same weight in the energy.** The two parities enter the flux (2.5) with identical weights and no cross term. Neither is “more important” by construction.

Figure 1 shows the $\ell = 2$, $m = \pm 2$ case, the dominant mode of a binary. For the even mode, $h_+ \propto (1 + \cos^2 \theta) \cos \Phi$ and $h_\times \propto 2 \cos \theta \sin \Phi$ with $\Phi = \omega u - 2\phi$: the familiar binary pattern. The odd mode has the two amplitudes exchanged. At the poles both are circularly polarized; at the equator the even mode is pure $+$ and the odd mode pure $\times$.

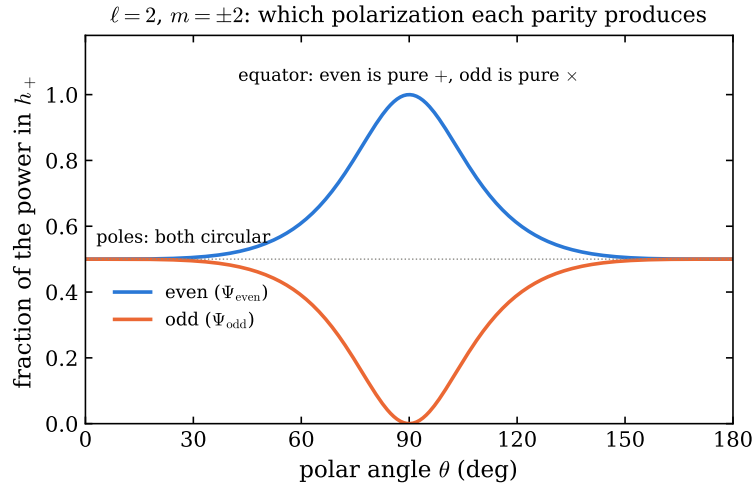


Figure 1: Fraction of the radiated power carried by h_+ as a function of the polar angle, for an $\ell = 2$, $m = \pm 2$ wave of either parity.

What we learned

h_+ and $h_\times$ are both built from both parities. The two parities are related by a 45° rotation of the polarization, and they carry energy with equal weight. The natural invariant basis is $(\Psi_{\text{even}}, \Psi_{\text{odd}})$, the E/B basis, not $(h_+, h_\times)$.

3 Can reflection turn $h_\times$ into h_+ ? A selection rule

Question

A wave hits the object and comes back. Can a wave that arrives as pure $h_\times$ come back with some h_+ ?

3.1 Parity is conserved

Key idea

The background (flat interior, shell, Schwarzschild exterior) is spherically symmetric *and* invariant under parity. The linearized equations therefore commute with rotations and with parity. Modes with different (ℓ, m) or different parity cannot be coupled, to all orders in linear theory and for any spherically symmetric object: black hole, shell, star.

In the language of the junction conditions, the even and odd systems separate completely: the odd sector involves only Ψ_{odd} , and the even sector only Ψ_{even} and the shell's even degrees of freedom. The response of the object to each (ℓ, m, parity) channel is a single complex number per frequency, the *reflection amplitude*

$$S_P(\omega) = \frac{B_{\text{out}}^P}{B_{\text{in}}^P}, \quad P \in \{\text{even}, \text{odd}\}, \quad (3.1)$$

where $\Psi_P \rightarrow B_{\text{in}}^P e^{-i\omega r_*} + B_{\text{out}}^P e^{+i\omega r_*}$. Both master functions are normalized in the same way relative to h by Eqs. (2.1)–(2.2), so the *ratio* $S_{\text{even}}/S_{\text{odd}}$ is physical.

3.2 The object as a polarizing mirror

For one (ℓ, m) , write the incident wave as a vector in the E/B basis. Reflection acts as a diagonal Jones matrix:

$$\begin{pmatrix} \Psi_{\text{even}} \\ \Psi_{\text{odd}} \end{pmatrix}_{\text{out}} = \begin{pmatrix} S_{\text{even}} & 0 \\ 0 & S_{\text{odd}} \end{pmatrix} \begin{pmatrix} \Psi_{\text{even}} \\ \Psi_{\text{odd}} \end{pmatrix}_{\text{in}} = S_{\text{odd}} \begin{pmatrix} \rho e^{i\Phi} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Psi_{\text{even}} \\ \Psi_{\text{odd}} \end{pmatrix}_{\text{in}}, \quad (3.2)$$

with the **retardance** $\Phi = \arg(S_{\text{even}}/S_{\text{odd}})$ and the **diattenuation** $\rho = |S_{\text{even}}/S_{\text{odd}}|$. This is exactly the optics of a birefringent mirror. It gives a precise answer to the question:

- A wave of *pure parity* keeps its parity. In particular, an axisymmetric ($m = 0$) pure- $h_\times$ wave is odd and **always comes back as pure $h_\times$** , with a different waveform. No object with spherical symmetry can convert it into h_+ .
- A wave containing *both* parities changes its polarization state, because the two components are reflected with different phases (and, if the object absorbs, different amplitudes). For a wave incident with equal E and B content (45° linear polarization in the E/B plane, e.g. an $m = 0$ wave with $h_+ = h_\times$), the fraction of the reflected power found in the orthogonal polarization is, for $\rho = 1$,

$$f_{\text{conv}} = \sin^2(\Phi/2). \quad (3.3)$$

The same expression gives the probability that a circularly polarized (helicity) mode comes back with the opposite helicity, because a helicity state is $\Psi_{\text{even}} \pm i\Psi_{\text{odd}}$.

The rest of the report computes $\Phi(\omega)$ for a black hole and for shells, and shows what it does to waveforms.

4 Reference case: a black hole is a pure retarder

Question

The Regge–Wheeler and Zerilli potentials are different, yet a Schwarzschild black hole has the same QNMs and the same reflectivity for both parities (isospectrality). Does that mean it treats h_+ and $h_\times$ identically?

Isospectrality follows from the Chandrasekhar transformation [11], which maps every solution of the Regge–Wheeler equation into a solution of the Zerilli equation:

$$\Psi_{\text{even}} = \left[\lambda_\ell + \frac{72M^2 f}{r(\mu r + 6M)} \right] \Psi_{\text{odd}} + 12M \frac{d\Psi_{\text{odd}}}{dr_*}, \quad \lambda_\ell = \mu(\mu + 2) = (\ell - 1)\ell(\ell + 1)(\ell + 2), \quad (4.1)$$

with $\mu = (\ell - 1)(\ell + 2)$. Take the black-hole scattering solution, which is purely ingoing at the horizon. On the horizon side $f \rightarrow 0$, so the map multiplies the ingoing wave by $\lambda_\ell - 12iM\omega$ and keeps the boundary condition. At infinity the extra term vanishes, and the map multiplies $e^{-i\omega r_*}$ by $\lambda_\ell - 12iM\omega$ and $e^{+i\omega r_*}$ by $\lambda_\ell + 12iM\omega$. Hence

$$\frac{S_{\text{even}}}{S_{\text{odd}}} = \frac{\lambda_\ell + 12iM\omega}{\lambda_\ell - 12iM\omega} \implies \rho = 1, \quad \Phi_{\text{BH}}(\omega) = 2 \arctan \frac{12M\omega}{(\ell - 1)\ell(\ell + 1)(\ell + 2)} \quad (4.2)$$

The black hole reflects the two parities with the same amplitude (isospectrality) but *not* with the same phase. It is a pure retarder. We computed S_{even} and S_{odd} independently, by integrating the Zerilli and Regge–Wheeler equations from the horizon, and recover Eq. (4.2) to better than 5×10^{-8} for $\ell = 2, 3, 4$ (Fig. 2). This also confirms that the numerical normalization of the two parities is consistent.

For $\ell = 2$ ($\lambda_2 = 24$) the retardance is 21° at the fundamental QNM frequency $\omega M = 0.374$ and grows to 90° at $\omega M = 2$. It is much smaller for higher ℓ ($\lambda_3 = 120$, $\lambda_4 = 360$). The effect comes entirely from the exterior Schwarzschild field: it would be the same for *any* compact object at frequencies where the wave does not reach its surface.

What we learned

Even a black hole is birefringent. Isospectrality means equal *moduli*, not equal *phases*. A wave that reflects off the potential barrier comes back with its E and B parts shifted by Φ_{BH} , so its polarization changes.

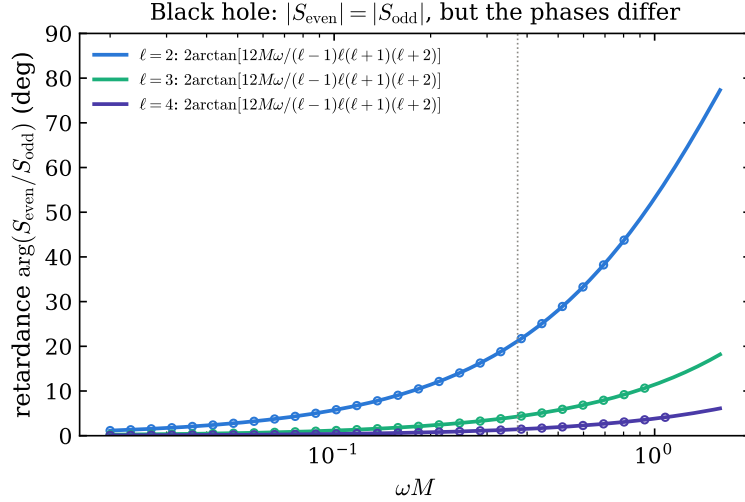


Figure 2: Black hole: retardance between the two parities. Lines: Eq. (4.2). Circles: independent numerical integration of the two master equations (shown where the reflectivity exceeds 10^{-6}).

5 The shell as a polarizing mirror

Question

The shell has no horizon and absorbs nothing, so $|S_{\text{even}}| = |S_{\text{odd}}| = 1$. How different are the phases, and when does the polarization change appreciably?

We computed $S_{\text{even}}(\omega)$ and $S_{\text{odd}}(\omega)$ for $\ell = 2$ and $R = 6M$, $3M$ and $2.2M$. The physical, regular interior was carried through the junction conditions and the exterior was integrated to large radius. Near the narrow resonances the frequency grid was refined adaptively (up to ~ 1400 frequencies per curve). Numerically $|S| = 1$ holds to 10^{-15} in both parities, as it must. Figures 3 and 3 show the retardance $\Phi = \arg(S_{\text{even}}/S_{\text{odd}})$ and the converted fraction $\sin^2(\Phi/2)$.

Low frequency: universal. For $\omega M \lesssim 0.1$ all shells follow the black-hole curve $\Phi = 2 \arctan(12M\omega/24)$. At $\omega M = 0.05$ we find $\Phi = 2.86^\circ$ for $R = 2.2M$ and $3M$ and 2.84° for $6M$, against 2.86° from Eq. (4.2). Long waves never reach the surface. They are reflected by the exterior Schwarzschild field, whose even/odd phase difference is fixed by the Chandrasekhar transformation. The structure of the object enters only at higher order (tidal response).

The matter resonance: complete conversion. The even sector has a long-lived matter mode, an oscillation of the shell’s fluid, at $\omega M = 0.0894 - 0.00005i$ ($R = 6M$), $0.1971 - 0.0003i$ ($3M$) and $0.1859 - 0.00003i$ ($2.2M$). The odd sector has nothing there. Across the resonance the phase of S_{even} winds by 2π while S_{odd} stays smooth. The retardance therefore sweeps through 180° and the conversion reaches $\sin^2(\Phi/2) = 1$: in a narrow band, of width $\sim |\text{Im } \omega|$, the shell *rotates the polarization of the reflected wave completely*. This is the clearest answer to “can $h_\times$ become h_+ ”: an $\ell = 2$, $m = \pm 2$ wave that arrives as pure $h_\times$ at the equator (pure odd) comes back as pure h_+ . But a wave arriving with equal E and B parts at the matter frequency comes back with its

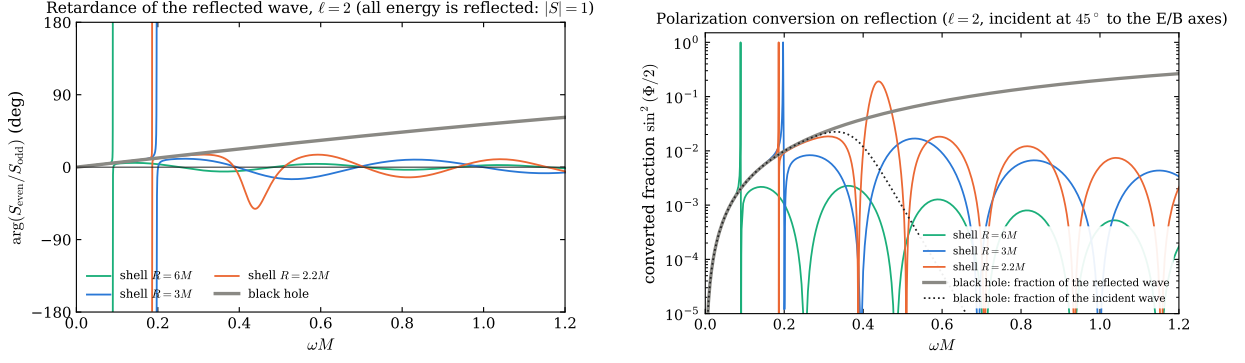


Figure 3: Left: retardance between the parities for three shells and the black hole ($\ell = 2$). Right: fraction of the reflected power that comes back in the orthogonal polarization, for a wave with equal E and B content. For the black hole the dotted line multiplies by the reflectivity, i.e. it is the fraction of the *incident* power.

polarization rotated by 90° , for instance $h_+ = h_\times$ returning as $h_+ = -h_\times$.

Intermediate frequencies: small but structured. Away from the matter resonance the retardance oscillates around zero. The two parities have cavity and trapped resonances at slightly different frequencies (e.g. $0.421 - 0.035i$ odd and $0.447 - 0.044i$ even at $R = 2.2M$). The frequency-averaged converted fraction on $0.1 < \omega M < 1$ is 0.08% ($R = 6M$), 0.7% ($3M$) and 1.7% ($2.2M$), with a maximum of 19% between the two trapped resonances of the $R = 2.2M$ shell. At high frequency the wave crosses the transparent shell and bounces off the centre, and both parities acquire the same phase: $\Phi \rightarrow 0$.

Contrast with the black hole. For a black hole the retardance keeps growing with frequency. At high frequency, however, almost nothing is reflected, so the converted fraction of the *incident* power (dotted curve) peaks at $\simeq 2\%$ near the barrier top and then falls. A shell returns everything, so its polarization signature can be much stronger (complete, at the matter resonance) but is confined to narrow bands.

What we learned

Parity is conserved, but polarization is not. A compact shell acts as a retarder whose retardance is universal at low frequency, oscillates at intermediate frequency, and sweeps through 180° at the even-only matter resonance, where the conversion is complete.

5.1 An absorbing interior: the parities are also reflected with different strength

If the interior is replaced by a perfect absorber (only an ingoing wave inside, as a horizon would do), the object also acquires a *diattenuation*: $\mathcal{R}_{\text{even}} \neq \mathcal{R}_{\text{odd}}$ (Fig. 4). The two reflectivities coincide at low and high frequency. Around the barrier top the even parity is reflected more strongly: by a factor of up to 3.4 ($R = 2.2M$) and 18 ($R = 6M$), and by a large factor at $R = 3M$ near $\omega M \approx 0.55$, where the odd reflectivity has an interference zero. The ratio never drops below 0.98. The black hole has $\mathcal{R}_{\text{even}} = \mathcal{R}_{\text{odd}}$ exactly. Such a diattenuating mirror turns an unpolarized wave into a partially polarized one, with an excess of the even (E -type) pattern.

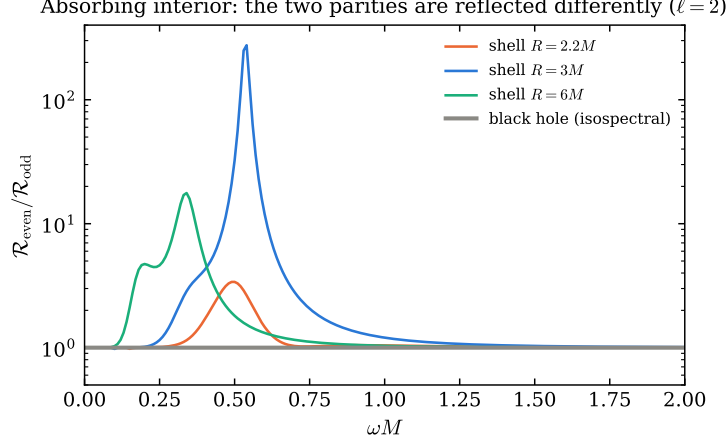


Figure 4: Ratio of the even to the odd reflectivity when the interior absorbs everything that crosses the shell.

6 Plane waves: how much helicity is reversed

Question

A distant source sends a circularly polarized plane wave onto the object. What fraction of the scattered radiation has the opposite helicity?

A plane wave of definite helicity has, for every $\ell \geq 2$, equal-magnitude even and odd components combined as $\Psi_{\text{even}} \pm i\Psi_{\text{odd}}$. After scattering, the helicity-preserving amplitude of partial wave ℓ is $(S_{\text{even}} + S_{\text{odd}})/2 - 1$ and the helicity-reversing amplitude is $(S_{\text{even}} - S_{\text{odd}})/2$. By orthogonality of the spin-weighted harmonics the helicity-reversing cross section is

$$\sigma_{\text{flip}}(\omega) = \frac{\pi}{\omega^2} \sum_{\ell \geq 2} (2\ell + 1) \frac{|S_{\text{even}}^\ell - S_{\text{odd}}^\ell|^2}{4}. \quad (6.1)$$

Unlike the total cross section, which diverges because of the long-range Newtonian potential, σ_{flip} is *finite*. The Newtonian phase is common to both parities and cancels in the difference.

Black hole, exactly. With Eq. (4.2), $|S_{\text{even}} - S_{\text{odd}}|^2/4 = \mathcal{R}_\ell \sin^2 \arctan(12M\omega/\lambda_\ell)$, where $\mathcal{R}_\ell$ is the reflectivity of partial wave ℓ . As $\omega \rightarrow 0$, $\mathcal{R}_\ell \rightarrow 1$ and

$$\sigma_{\text{flip}} \rightarrow 144\pi M^2 \sum_{\ell \geq 2} \frac{2\ell + 1}{[(\ell - 1)\ell(\ell + 1)(\ell + 2)]^2} = 144\pi M^2 \cdot \frac{1}{108} = \frac{4\pi}{3} M^2. \quad (6.2)$$

The sum can be done in closed form (symbolic summation gives exactly $1/108$). The result coincides with the angular integral of the helicity-reversing part, $M^2 \sin^4(\theta/2)$, of the classic long-wavelength cross section $d\sigma/d\Omega = M^2 [\cos^8(\theta/2) + \sin^8(\theta/2)] / \sin^4(\theta/2)$ [15–18]: $\int M^2 \sin^4(\theta/2) d\Omega = 4\pi M^2/3$. The partial-wave formalism, the relative normalization of the two parities and the Chandrasekhar phase are therefore all consistent with the known result.

Shells. Evaluating Eq. (6.1) numerically (ℓ up to $\sim 2.2\omega R + 7$, last term below 5×10^{-4} of the sum) gives Fig. 5. At $\omega M = 0.02$ all objects give $\sigma_{\text{flip}} = 4.188M^2$, the universal value (6.2). The helicity flip at long wavelengths depends only on the mass. At higher frequency the curves separate. The dilute shell ($R = 6M$) flips less than a black hole around $\omega M \sim 0.1$ – 0.5 , because its partial waves pass through the shell and the centre almost symmetrically. The compact shells flip more than the black hole for $\omega M \gtrsim 0.4$, where the black hole absorbs almost everything, and the

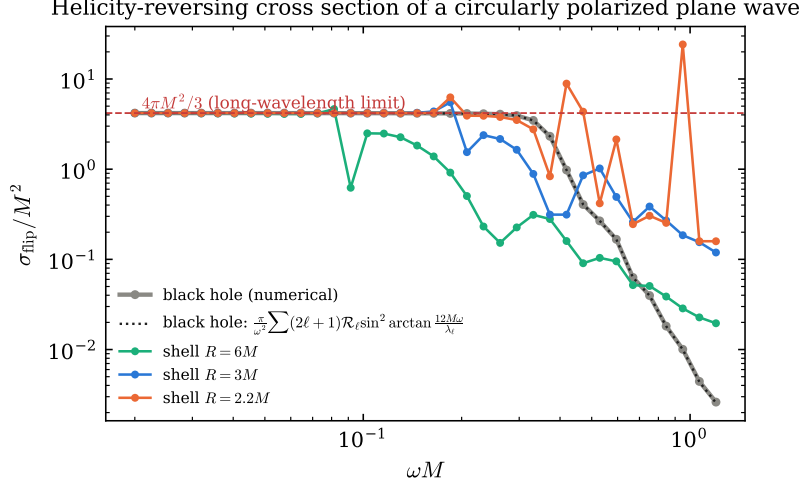


Figure 5: Helicity-reversing cross section, Eq. (6.1), for the black hole (numerical, and the closed form with numerical reflectivities) and three shells. Dashed: the universal long-wavelength value $4\pi M^2/3$.

$R = 2.2M$ shell shows sharp peaks (up to $\sim 24M^2$ on our grid) at the trapped resonances of the higher multipoles. These peaks are only partly resolved by the frequency grid.

What we learned

Any spherical compact object reverses the helicity of long gravitational waves with cross section $4\pi M^2/3$, independent of its structure. Structure-dependent polarization signatures appear at $\omega \sim 1/R$ and are strongest near resonances.

7 Waveforms: the same incident h_+ pulse in the two parities

Question

What does a detector see if the same pulse arrives once carried by the even master function and once by the odd one?

Set-up. For a single (ℓ, m) the angular factor in Eq. (2.4) is the same for the incident and the reflected wave. So h_+ and $h_\times$ at any fixed direction are that angular factor times $\Psi_{\text{in}}(v)$ before the reflection and $\Psi_{\text{out}}(u)$ after it. For instance, for $m = 0$ the even channel is literally “incident h_+ , reflected h_+ ”, and the odd channel is the same with $h_\times$. We therefore compare the reflected $\Psi_{\text{out}}(u)$ of the two parities for the same incident pulse,

$$\Psi_{\text{in}}(v) = e^{-v^2/2\sigma^2} \cos(\omega_0 v), \quad \sigma = 8M, \quad \omega_0 = 0.5/M, \quad v = t + r_*, \quad (7.1)$$

by Fourier synthesis, $\Psi_{\text{out}}(u) = \frac{1}{2\pi} \int S(\omega) \tilde{\Psi}_{\text{in}}(\omega) e^{-i\omega u} d\omega$, with $u = t - r_*$. The integral is done with a Filon-type quadrature on the adaptive frequency grid, so the narrow resonances are resolved.

Causality and the instability. The fluid shell has an unstable even-parity matter mode at $\omega = i\gamma$. There, S_{even} has a pole in the upper half plane, and the real-axis integral is then *not* the causal response. The causal (retarded) response is obtained by moving the contour above the pole:

$$\Psi_{\text{out}}^{\text{causal}}(u) = \Psi_{\text{out}}^{\text{real axis}}(u) - i \text{Res}_{\omega=i\gamma} [S_{\text{even}} \tilde{\Psi}_{\text{in}}] e^{\gamma u}. \quad (7.2)$$

The residue must be computed directly. For $\text{Im} \omega > 0$ the outgoing solution decays with r , so we project the exterior solution onto the in/out asymptotic solutions at a moderate radius. This reproduces the real-axis S to 10^{-8} , and Newton iteration on $B_{\text{in}} = 0$ returns the pole exactly at

the known unstable frequency: $0.1332058i$ ($R = 3M$), $0.2442921i$ ($2.2M$), $0.0401379i$ ($6M$). The residues, stable to three digits against the projection radius, are tiny: $7.5 \times 10^{-5}i$, $8.2 \times 10^{-6}i$ and $1.1 \times 10^{-5}i$. The unstable mode lives in the shell's matter and is only weakly coupled to waves coming from infinity. For this pulse its amplitude at $u = 0$ is 3.9×10^{-7} ($R = 3M$), 1.4×10^{-8} ($2.2M$) and 2.3×10^{-8} ($6M$). It reaches the amplitude of the incident pulse at $u \approx 111M$, $74M$ and $438M$ respectively.

Check. For the odd channel the result was compared with a direct time-domain (leapfrog) evolution of $\partial_t^2 \Psi = \partial_x^2 \Psi - V\Psi - \Delta \delta(x - x_R)\Psi$ that shares no code with the frequency-domain calculation. With source and detector far from the object ($x = 450M$ and $350M$), and the $O(1/\omega r)$ near-zone factors of the in/out solutions included, the two agree to 0.4% ($R = 3M$) and 0.9% ($2.2M$) of the peak (Fig. 7).

What the waveforms show (Fig. 6).

- **Black hole.** The two waveforms are almost identical. The even one is the odd one passed through the all-pass filter $(\lambda_\ell - 12M\partial_u)/(\lambda_\ell + 12M\partial_u)$, the time-domain form of Eq. (4.2). This gives a delay of $\simeq 24M/\lambda_\ell = 1M$ for $\ell = 2$ at low frequency. Both ring down at the same QNM.
- **Shell at $3M$.** The prompt reflection is again nearly the same in both parities (the retardance is only $\sim 10^\circ$ in this band). The difference is at late times: the even channel keeps a weak ringing at the matter frequency $\omega M = 0.197$, with a decay time $\sim 3000M$. On top of it the unstable mode grows as $e^{0.133u/M}$ and dominates after $\sim 110M$. The odd channel has neither.
- **Shell at $2.2M$.** Both parities ring in their trapped modes, at slightly different frequencies (0.421 odd, 0.447 even) and dampings (0.035 vs 0.044). The two waveforms dephase visibly after a few cycles, and the unstable mode takes over the even channel after $\sim 74M$.
- **Shell at $6M$.** The prompt reflections are indistinguishable. The even channel carries a long matter ringing at $\omega M = 0.089$ at the 10^{-5} level and blows up after $\sim 440M$.

Polarization in the time domain. For a pulse with equal E and B content, the part that comes back with the orthogonal polarization is $(\Psi_{\text{out}}^{\text{even}} - \Psi_{\text{out}}^{\text{odd}})/2$ (Fig. 7). It carries 3.7% of the reflected energy for the black hole and 1.0% for the $R = 3M$ shell (stable part), concentrated at the edges of the pulse, where the phase difference acts. In the shell case the rotated component then persists at late times, carried by the even-only matter ringing.

What we learned

Plotted as $h_+(t)$, the even and odd reflections of the same pulse look alike during the prompt response. Their difference grows with time: in the dephasing of the trapped modes, in the even-only matter ringing, and above all in the even-only instability of the static shell.

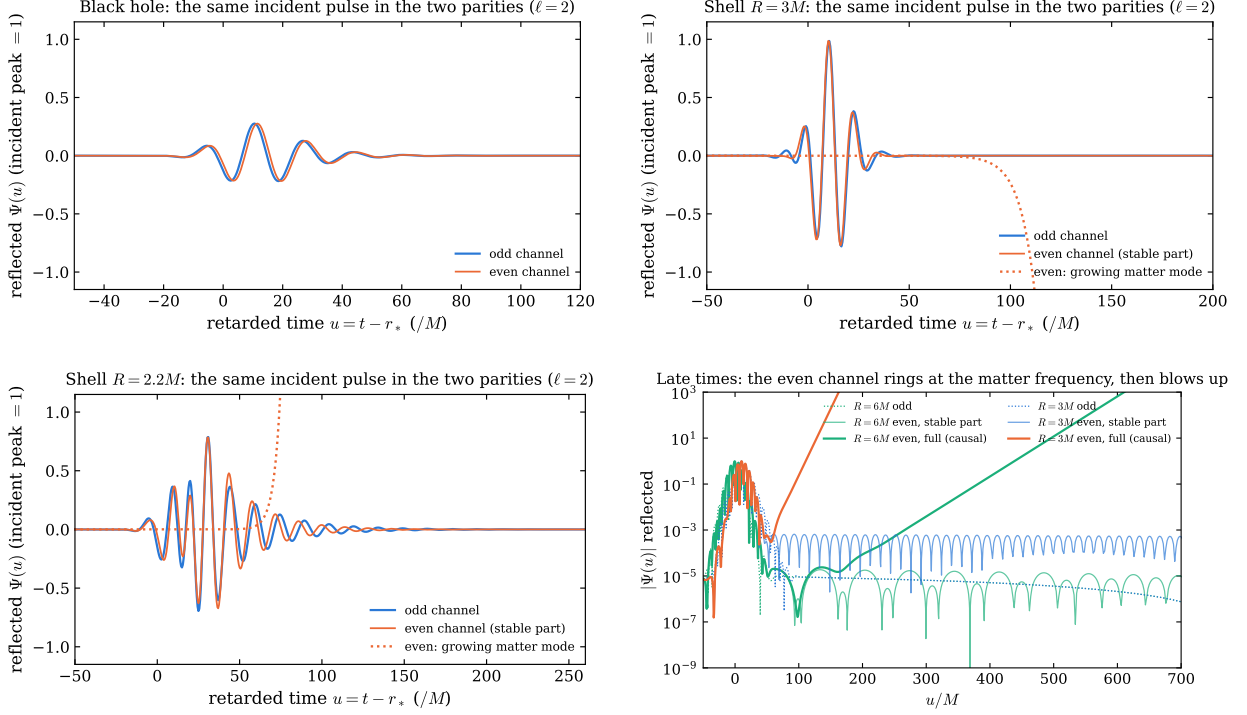


Figure 6: Reflected waveforms of the same incident pulse, carried by the odd (blue) or even (orange) master function. Top left: black hole. Top right: shell at $3M$. Bottom left: shell at $2.2M$. Dotted: the growing matter mode of the even channel, added causally. Bottom right: long-time view (log scale) for $R = 6M$ and $3M$. The even channel keeps ringing at the matter frequency and is eventually overtaken by the unstable mode; the odd channel just decays. The floor at $\sim 10^{-5}$ is numerical.

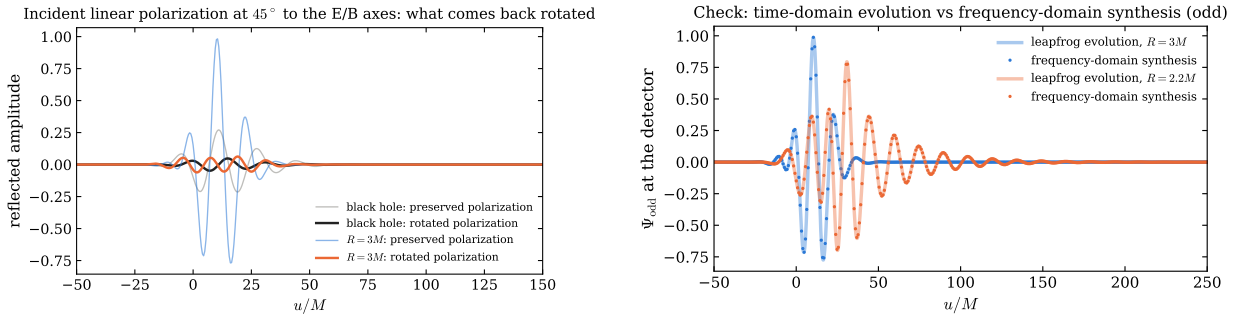


Figure 7: Left: preserved and rotated parts of a reflected pulse with equal E and B content. Right: check of the frequency-domain synthesis against an independent time-domain evolution (odd channel).

8 So, is one parity more important?

Question

For a harder problem one would like to know whether one parity can be neglected. What do the results say?

The answer depends on which of three questions is being asked.

1. Kinematics: no. The two parities carry energy with identical weights, Eq. (2.5), and they are exchanged by a 45° rotation of the polarization. A plane gravitational wave contains equal amounts of both: its partial-wave expansion has $\Psi_{\text{odd}}^{\ell,\pm 2} = \pm i \Psi_{\text{even}}^{\ell,\pm 2}$ for every ℓ . For *scattering of an external wave* both parities are equally excited, and dropping one loses half of the signal and all of the polarization physics of Secs. 5–7.

2. Response of the object: comparable in energy, different in phase and in physics.

- For a non-absorbing object (the shell) $|S_{\text{even}}| = |S_{\text{odd}}| = 1$. Both parities return exactly the energy they bring. For a black hole the moduli are equal by isospectrality. With an absorbing (“open”) interior the two reflectivities differ only at intermediate frequencies, near the barrier top, and coincide at low and high frequency.
- What differs is the *phase* and therefore the *time structure*: the retardance, the ringing frequencies and their damping (Secs. 5, 7).
- Qualitatively, only the even sector *moves the shell and couples to its matter*: radial displacement, compression and the equation of state all enter through Ψ_{even} . The odd sector sees a universal δ -barrier, $[\partial_x \Psi] = \Delta \Psi$ with $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$, which does not depend on the frequency or on the equation of state. The matter resonances, their long ringing, and the instability of the static shell exist *only* in the even sector.

3. Sources: even dominates for slow sources. For a source with internal velocities $v \ll 1$, the even $\ell = 2$ mode is the mass quadrupole and the odd $\ell = 2$ mode is the current quadrupole [13]. The latter is suppressed by one power of v in amplitude. For a circular binary of masses m_1, m_2 and total mass m , the post-Newtonian flux gives [14]

$$\frac{\dot{E}_{\text{odd}, \ell=2}}{\dot{E}_{\text{even}, \ell=2}} \simeq \frac{1}{36} \left(\frac{m_1 - m_2}{m} \right)^2 v^2, \quad (8.1)$$

of the same order as the even $\ell = 3$ (mass-octupole) contribution. A radially infalling or head-on source is axisymmetric and polar and excites *only* the even sector. Odd parity is dominant only for sources driven by currents: rotation, toroidal flows, or axial (“ w -mode” or r -mode-like) oscillations.

Key idea

Advice for a harder problem. (i) If the problem is the scattering of an external wave, keep both parities. They carry equal energy, and the polarization of the outgoing wave is set by their relative phase. (ii) If the waves are generated by slow matter motion, the even sector dominates and the odd sector is an $O(v^2)$ correction in power. (iii) In either case the odd sector is the *simpler* and more universal one: for a thin shell it reduces to a δ -barrier. It is a good first test of the code, while the even sector contains the matter physics and the possible instabilities.

9 Conclusions

- 1. Polarizations vs parities.** $h_+ - ih_\times = \frac{1}{2r} \sum \sqrt{(\ell+2)!/(\ell-2)!} (\Psi_{\text{even}} - i\Psi_{\text{odd}}) {}_{-2}Y^{\ell m}$. The even and odd master functions are E - and B -type patterns related by a 45° rotation of the

polarization. Each contributes to both h_+ and $h_\times$; only for $m = 0$ is even = + and odd = $\times$.

2. **Selection rule.** Spherical symmetry and parity forbid even–odd mixing for any spherical object. A wave of pure parity keeps its parity: an axisymmetric $h_\times$ wave cannot come back as h_+ .
3. **Birefringent mirror.** A wave containing both parities does change polarization, because $S_{\text{even}} \neq S_{\text{odd}}$. A black hole is a pure retarder with $\Phi = 2 \arctan[12M\omega/(\ell - 1)\ell(\ell + 1)(\ell + 2)]$, exactly, from the Chandrasekhar transformation (verified numerically to 5×10^{-8}).
4. **Universal long-wavelength helicity flip.** $\sigma_{\text{flip}} \rightarrow 4\pi M^2/3$ for any spherical compact object. We derived this from a partial-wave sum that equals $1/108$ exactly, and it agrees with the classic Schwarzschild cross section.
5. **Shell-specific polarization effects.** Complete polarization conversion occurs in narrow bands at the even-only matter resonance. Conversions of a few per cent (up to 19% for $R = 2.2M$) occur between the even and odd trapped resonances. With an absorbing interior, the even parity is reflected up to an order of magnitude more strongly near the barrier top.
6. **Waveforms.** The same incident pulse gives nearly identical prompt reflections in the two parities. The even channel differs at late times: dephased trapped ringing, a long ringing at the matter frequency, and the growing unstable mode. The latter is weakly excited (residue $\sim 10^{-5}$) but dominates after $\sim 74\text{--}440M$, depending on R .
7. **Which parity matters.** In energy neither: the weights are equal, a plane wave contains both equally, and the reflected energies are equal for a non-absorbing object. In physics, the even parity carries all the coupling to the matter and is the dominant emission channel of slow sources. The odd parity is simpler and universal, a δ -barrier.

Methods and checks. Reflection amplitudes come from the exact junction conditions of the shell (odd: analytic; even: 2×2 matrix from the linearized Israel equations), a regular flat interior, and high-accuracy integration of the Regge–Wheeler and Zerilli equations. The checks are:

- the black-hole phase ratio against the Chandrasekhar formula (5×10^{-8});
- $|S| = 1$ for the shells (10^{-15});
- the universal low-frequency retardance and helicity-flip cross section;
- the unstable-pole position against the known matter mode (seven digits);
- the odd-channel waveforms against an independent time-domain evolution ($< 1\%$).

Code. `code/parity_study.py`: all computations, cached in `data/`. `code/residue.py`: residue of S_{even} at the unstable pole. `code/make_figures_r4.py`: figures; numbers quoted in the text are in `code/numbers_r4.json`.

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